Neural Ensemble Decoding of Rat's Motor Cortex by Kalman Filter and Optimal Linear Estimation

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Abstract

In this paper, rats were trained to press a lever over a threshold to get water as rewards, and neural ensemble activities in primary motor cortex (MI) and pressure signal of the lever were recorded synchronously. Meanwhile, two algorithms, Kalman filter (KF) and Optimal Linear Estimation (OLE), were used to decode neural ensemble activities around the pressing events. After training, the pressure values were extracted from the neural signals by those decoders, and the performances were compared by Correlation Coefficient (CC) and Mean Square Error (MSE) with pressure values recorded from the lever. KF was better than OLE when the bin size < 100ms with CC > 0.7, but it was not applicable in this experiment when the bin size >100ms for excessively smoothed. And OLE performs better when bin size >100ms for diminishing the randomicity of neural firing, which might imply that KF is more applicable in real-time system.

Keywords: Brain-Computer Interface; neural decoding; Kalman Filter; Optimal Linear Estimation.

1. Introduction

The technology of Brain-Computer Interface (BCI) has been established and developed in last decades, in order to decode cortical neural activity into control commands of devices to rehabilitate or extend physiological functions. And the decoding of movement information according to cortex's neural activity has been a focus of the research of BCI[1-3]. Preceding studies prove that the neural ensemble activity of motor cortex is distributed but not centralized[4]. Though traditional recording technology of single electrode could record electrical activity of one or several neurons, it hardly demonstrates the synergic characteristic of neural firing in population-level. In recent years, the development of techniques for simultaneous population-level neural recording enables the research of distributed coding in motor cortex [5-8].

Recently, lots of methods have been developed for decoding motor cortex activity, including the population-vector algorithm, optimal linear estimation, artificial neural networks and Bayesian methods. Population vector (PV) algorithm has been a successful neural ensemble decoding [9, 10] method, in which each neuron's activity is characterized by its preferred direction and firing rate. PV is an optimal solution when the tuning functions are linear and the preferred directions are uniformly distributed[11]. Optimal linear estimation (OLE) is used when the preferred directions are not uniformly distributed [12], which uses least squares. It is regarded that OLE is a more general method and PV method is a special case of OLE. But both of them hardly consider the dynamic relation between neural activity and movement, they assume that the firing rates are linearly related to the underlying movement parameters. Artificial Neural Networks (ANN) has also been proved successful in neural ensemble decoding in motor cortex, but the results seems not very different from PV and OLE, and research has demonstrated that artificial neural networks actually encode a linear model as the PV method[13]. The Kalman filter (KF) provides an efficient recursive algorithm to Bayesian approach when the likelihood and prior models are linear and gaussian. It solved the problem of both ANN and PV that they lack both a clear probabilistic model and a model of the temporal kinematics, and provide no estimate of uncertainty and may be difficult to extend the analysis of more complex temporal movement patterns[14].

In this paper, neural ensemble activity in rat's motor cortex was decoded by KF and OLE. In the experiment, a 2×8 microelectrode array was implanted in rat's motor cortex to record neural spiking signals, and water reward was given when rat pressed a lever by its pre-limb. Spikes were extracted and sorted from multi-channel neural signals and the firing rate of each neuron was computed, And the simultaneously recorded pressure signal of the lever was also discretized. The process of KF decoding was as follows: Firstly, a length of training data that included a lot of pressing events, both neural firing and pressure signal, was used to train the parameters in KF model. Secondly, the pressure of lever was predicted given current neural activity and the linear dynamic relationship between current and future pressure states. Similar to KF, the OLE was also used in two steps: train and decode. At last, the performances of

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two decoders were compared in a range of bin size (10 ms,20 ms,…,300ms) by correlation coefficient and mean square error with detected pressure.

2. Methods

2.1. Experiment Environments

All rats were trained to perform conditional operant task in which a lever was pressed down over a threshold and then animal were rewarded by water, figure 1.a. During the training, the rats were kept on water restriction and maintained at a level in which 12ml water was provided per day for a rat. The average training time was 5–7 days (30 min training sessions per days). All rats were trained to achieve a success rate of >75% before they were implanted.

2.2. Multi-Electrodes Implant Surgery

Approval for all experimental procedures was obtained from the Animal Care Committee at Zhejiang University, China, that follows the Guide for the Care and Use of Laboratory Animals (China Ministry of Health).

Methods for surgery, multineuron recording have been published[15]. Briefly, custom-made 2×8 channel microwire electrodes arrays (diameter = 35µm, Stablohm 675, California Fine Wire,) were chronically implanted with stereotaxic techniques into the forepaw region of the forelimb primary motor cortex in three male, Sprague–Dawley rats (approximately 280g). The target depth was chosen to be 1.2mm. The approximate depth of layer V is in the range of 1.1mm–1.8mm beneath the pia in the cerebral cortex and contains cell bodies that project directly to the spinal cord to activate peripheral motor neurons.

A typical recovery time of 5–7 days was needed before routine measurements began. Analog signals were filtered and amplified using Cerebus 128TM (Cyberkinetics Inc, US), and single neurons were discriminated on each electrode using custom-made software with PCA and K-Means.

2.3 Kalman Filter

In this paper, Kalman filter (KF) was used to estimate rat's pre-limb pressure from its neural activity. As the most basic of the Bayesian approaches, KF provides an efficient recursive algorithm to optimally estimate the posterior probability when the likelihood and prior models are linear and Gaussian [14]. So, in our example, the method assumes that the relationship between the pre-limb motion and the neural activity is linear, and the noise in the neural ensemble firing obeys Gaussian distribution.

In the KF model, rat's pre-limb pressure was defined as p(n) in the decoding process at time $t_n=n\Delta t$, where Δt was bin size in our experiments. The neural ensemble activities of N neurons were defined as $f_i(n)$ which represents the firing number of the *i*th neuron in *n*th Δt, and N represents the number of recorded neurons. Hence, the state vector was defined as:

$$x(n) = \left[p(n) \quad f_1(n) \cdots f_N(n) \right]^T \tag{1}$$

The KF model contains two parts in our work: Firstly, encoding neural activity according to outside pressure state by a segment of training data; Secondly, decoding outside pressure state from neural ensemble firing using the firstly obtained linear relation.

In the encoding part, the states were assumed to propagate in time according to the following linear dynamic equation, so-called generative model:

$$x(n+1) = A_n x(n) + q_n \tag{2}$$

demonstrating that the pressure state at time n+1 was linearly related with the pressure state at time n, where A_n is a N×N square matrix and q_n denotes the a zero-mean noise term with covariance Q_n ($q_n \sim N(0, Q_n)$).

And the pressure state output can be generated from another equation:

$$y(n) = H_n x(n) + v_n \tag{3}$$

where v_n was the zero mean measurement noise with covariance V_n and $y(n)$ denoted the output (neuron ensemble activity). From the definition of $x(n)$ and $y(n)$, we got $H_n = \left[0 \quad I_{31\times 31}\right]$ and $v_n = 0$. And then, we computed parameters in KF filter from training data, which contained both neural ensemble activity and outside pressure state, with the least squares estimation by the following equation:

$$A = \arg\min_A \sum_{n=1}^{L-1} \left\| x(n+1) - Ax(n) \right\|^2 \tag{4}$$

The optimal solutions of A and Q in our model were as follows:

$$A = X_1 X_0^T \cdot inv(X_1 X_1^T) \tag{5}$$

$$Q = \frac{(X_1 - AX_0)(X_1 - AX_0)^T}{L-1} \tag{6}$$

where $inv(\cdot)$ and $(\cdot)^T$ denoted inverse operation and transpose operation of a matrix respectively. X_1 was a train of states and X_0 was another train with same length and after X_1 by one time step.

In the decoding part, following equations were used to estimate pressure state in each step:

$$P'(n+1) = AP(n)A^T + Q \tag{7}$$

$$K(n+1) = P'(n+1)H^T inv(HP'(n+1)H^T) \tag{8}$$

$$\tilde{x}(n+1) = A\tilde{x}(n) + K(n+1)(Y(n+1) - HA\tilde{x}(n)) \tag{9}$$

$$P(n+1) = (I - K(n+1)H)P'(n+1) \tag{10}$$

where $K(n)$ represented obvserver gain and $P(n)$ represented error variance. Equation (8) was used to update the KF gain, which produced a state estimate that minimizes the mean squared error of the reconstruction. Equation (7) and (10) were used to update the state error covariance in each step. Using the current state estimate $\tilde{x}(n)$ and neural activity in the next bin $Y(n+1)$, the next state value was estimated by Equation (9).

2.4 Optimal Linear Estimation

In this paper, Optimal Linear Estimation (OLE) was used to decode rat's neural ensemble activity as well. As a generalization of Population Vector algorithm, uniform preferred directions were not required in OLE. In our case, the OLE estimated by

$$p(n+\tau) = y(n)\beta \tag{11}$$

where $p(n+\tau)$ represented the estimated pressure value by neural ensemble activity $y(n)$ in time n, with time lag τ. The coefficient matrix can be obtained by minimizing mean squared error as Equation (12)

$$\beta = inv(Y^T Y)Y^T P \tag{12}$$

where Y and P were neural ensemble activity and pressure signal of a length of training data.

3. Results

3.1 Surgery and signal recording

Five rats were trained to obtain water by pressing a lever, and three of them (S9-01, S9-02, S9-03) were implanted 2×8 microelectrode array later. After a week's period of restoring from surgery and another week of training, all of three rats could obtain water by pressing lever adroitly and repeatedly with a period, about 3-5 seconds. The neural signals of these rats got stable gradually in two weeks from surgery, and after signal pre-processing, spike detection and spike sorting algorithm, 18 (S9-01), 27 (S9-02) and 31 (S9-03) spikes were extracted from them. The sorts of spike that happen rarely and are irrelevant with the pressing event are not taken into account and were not considered in later analysis. Data of simultaneously recorded neural signal and pressure signal with length of 410 seconds (S9-03), containing 77 pressing events, were used in this paper. The data in the first 100 seconds with 15 pressing events were used to train the KF and the OLE model as the process of encoding, and the other 310 seconds data were used to test the model as the process of decoding.

To find the time lag between the neural ensemble activity and the pressure state around the pressing events, the pressure signal was rearranged together by a threshold in the training data. Dashed line in Figure 2.a represented the threshold value. Then, the averaged pressure signal around pressing event was also found, as the black thick line in Figure 2.a. Similarly, the averaged neuronal firing rates of 31 neurons around the pressing events were also computed, as the 31 thin lines in Figure 2.b. The value of each point in the line was assigned as neuron index (1,2,…,31) plus the averaged spike firing number in the bin. After that, we needed to find the firing trend of all neurons around the pressing events. Principal components analysis was employed, and the first principal component was used to represent the firing trend of all neurons, as black thick line in Figure 2.b. By computing the cross-correlation function between the averaged pressure signal and the first principal component of 31 averaged neural firing, the time lag between the neural ensemble activity and the pressure state was found about 300ms.

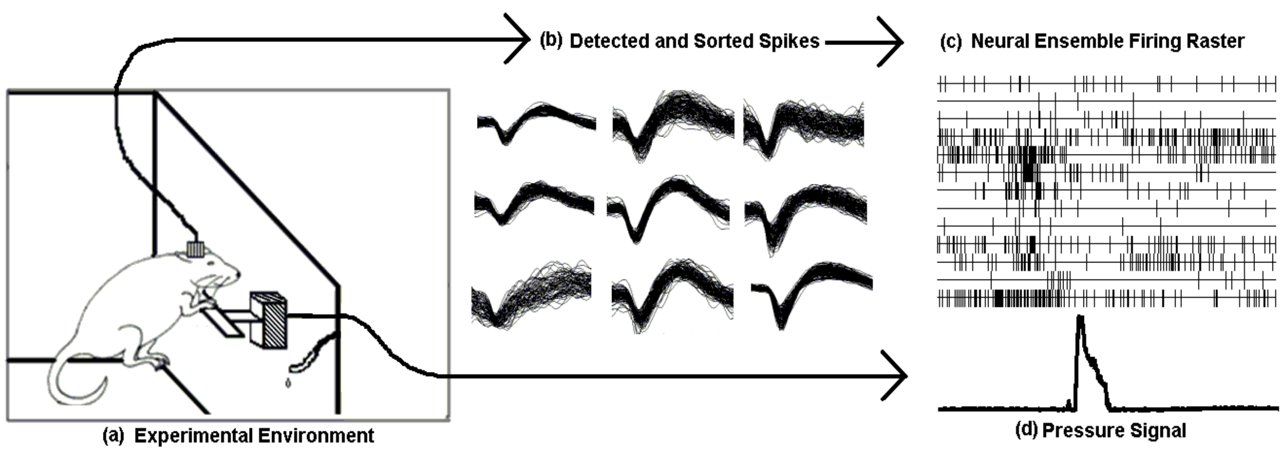

Figure 1. Experimental environment and simultaneously recorded neural signal and pressure signal
(a) Experimental environment. Water was rewarded when rat pressed the lever. (b) Part of extracted spikes in rat S9-03 by 2×8 microelectrode array. (c) Raster of neuronal ensemble firing in 5 seconds, each vertical short line in the figure stood for a spike firing. (d) Pressure value of the lever from pressure sensor. An abrupt change of the pressure value indicates a pressing event of rat. When the rat did not press the lever, the pressure value rested in the base line.

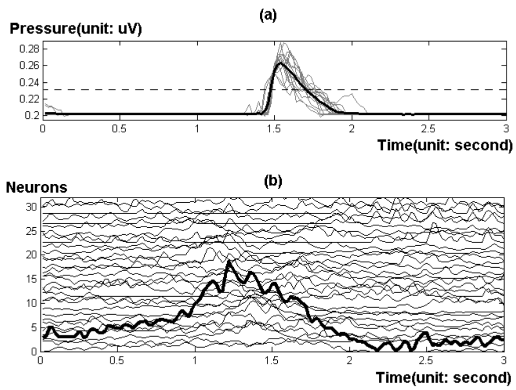

Figure 2. Rearranged pressure data and neural ensemble activity around pressing event

(a) Pressure data around 15 trails of pressing events. The gray thin lines are pressure signals around each 15 pressing events and the black thick line is the average of 15 trails. (b) 31 thin lines represent averaged neural activity of 15 trails of 31 neurons binned in 50ms, and the values of them are the normalized spike counts in the bin. The thick line is the first principal component of 31 neurons' activity.

After that, the KF model was trained by the 100 seconds continuous training data. In this process, the encoding matrix A and Q were obtained by Equation (5) and (6) with time lag 300ms. And the update process was applied to decode neural ensemble activity into pressure signal then. The Mean Square Error (MSE) of output of the KF was computed, as well as the Correlation Coefficient (CC) with actually detected pressure value of test data. Similarly, the OLE decoder was trained by the same period of data, and the error level of OLE was also found.

A data of 310 seconds were got through these decoders to test the performances of them. When the bin width was 50ms, parts results of these two methods were shown as Figure 3. The output of KF was smooth and exhibit clear peak when the pressing event was happened, but the waveform was in disorder between the pressing events. There were evident peaks in the OLE output during the pressing events as well, but the whole output waveform was corrupted by high frequency noise. The CC between the KF output and the real pressure signal was 0.7129, and the correlation coefficient between the OLE output and the real pressure signal was 0.1077. The MSE of KF output was 0.5740 and the value was 4.0391 for OLE output. Both parameters indicated that KF decoder performed better than OLE when the bin size was 50ms.

After that, the decoding performances of these two methods were tested in a range of bin size (10ms, 20ms, …, 300ms). Correlation coefficient and MSE were found in Figure 4. The performance of KF was better than OLE when the bin size was less than 100ms, with higher CC and lower MSE. When the bin size was larger than 100ms, the performance of KF deteriorated rapidly and retained in a certain level in the range of 100ms~300ms. On the contrary, the performance of OLE improved gradually by the increasing of bin size, with gradually increased CC and decreased MSE.

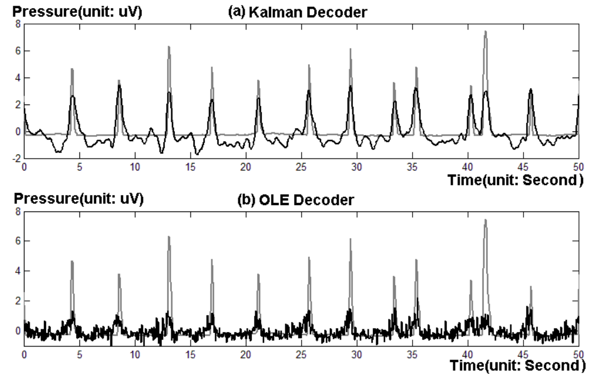

Figure 3. Decoding results of Kalman filter Decoder and OLE Decoder

(a) Result of KF. The gray line represents KF output, and the black line represents actual pressure value detected by pressure sensor. (b) Result of OLE. The gray line represents OLE output, and the black line represents actual pressure value.

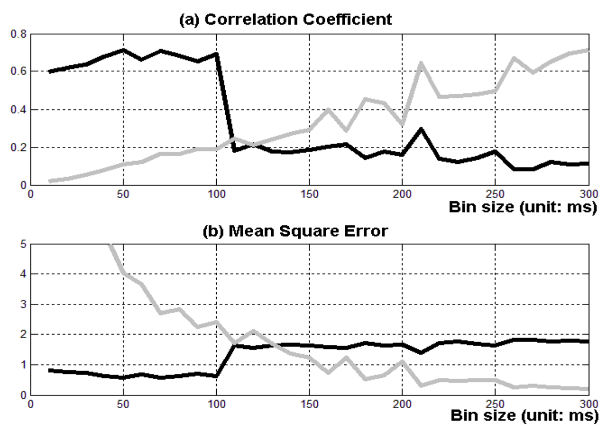

Figure 4. Performances of decoders in a range of bin size

(a) Correlation coefficient between decoder outputs and real detected pressure. The black line represented the performance of KF, and the gray line represented the performance of OLE. (b) Mean Square Error between decoder outputs real detected pressure. The black line denoted the MSE of KF, and the gray line denoted the MSE of OLE.

4. Discussion

In this paper, rats were trained to obtain a drop of water by pressing lever. The neural ensemble activities in primary motor cortex and the pressure signals were recorded synchronously. KF and OLE were used to decode neural ensemble activities in MI around the pressing events. In the beginning, a group of training data, containing both neural firing information and pressure value, was used to train the decoders. After that, decoded pressure was generated by those decoders in a range of bin size. The results showed that KF had better performance than OLE when the bin size was smaller than 100ms, with higher correlation coefficient and lower mean square error when decoded pressure signals compared with actually recorded ones. But the result was opposite when the bin size was larger than 100ms.

OLE assumed that each neuron was considered independently in its contribution to the movement, as well as the neural firing was linearly combined to generate movement. Results in this paper indicated that OLE required large spike firing bin to diminish the randomicity of neural firing. It should be noted that, as a linear estimation method, OLE was usually applicable to stationary processes, but not to the activity of neural ensemble which was more like a nonstationary random process actually. So, when the randomicity of neural firing was eliminated to a certain extent in the company of enlarging the bin size, the performance of OLE was enhanced.

Though performance of KF seemed deteriorated when the bin size was larger than 100ms, by the measurements of CC and MSE, we cannot conclude that KF performs worse than OLE in large bin size range abruptly. By analyzing the waveform of the KF outputs in large bin size (100ms~300ms), we found that the peaks were still very clear when the pressing events were happening and that the outputs seemed to be excessively smoothed. Since the pressing event in our experiment was usually with specified duration and the KF output would be excessively smoothed with large bin size, the width of KF peak would be wider than the real pressing event, and the CC would be decreased, as well as increased MSE.

As a result, in our experiment, KF performed better than OLE with higher CC and smaller MSE for smaller bin size (10ms~100ms). And when the bin size was larger than 100ms, the KF was no longer applicable in our experiment. In fact, when we applied these methods in real-time experiment, a larger bin usually performs inferior instantaneity. Our future work is going to control devices using decoded neural signal by these methods.

5. References

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